PORJECT REPORT

ON

Budget Tracking &
Reporting System

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Introduction-

Managing money is necessity in now a day life, since we don’t know when the financial crises come. that’s why I have create a simple but effective python program that will help the individual person or business to tracking their income and expenses effectively because manual tracking is often inaccurate or tedious job to do as well.

This project will help the user to make their own personal budget report and allows the user to enter there income and expenses, after which it calculates the total spending money and the remaining balance that’s was displays in a proper tabular format.

Problem Statement-

People these day’s often feel difficulties in managing there expenses and control there budget of the month.

The goal is to create a program that:

1. Accept the income.
2. Accept multiple spending to enter by user.
3. Calculate the finial summary for the report.
4. Make a table list the above.

Functional Requirements-

1.Progeam must accept the income from the user.

2.progam can take multiple input related to expenses.

3.program must calculate the total expenses.

4.program must calculate the remaining balance.

5.program must create a tabular form that show the above information in clean from.

6.program should stop taking the input when the user enter “done”.

7.program should identify that the input numerical in the amount should not be invalid.

Non-Functional Requirements-

1. The output should be simple and easy to navigate through.
2. The code must be short and have else time complexity for it to work most efficiently and generate the output instantly.
3. The calculation done by the code must give the optimal and correct answer when calculating.
4. The program should handle individual numerical separately for less strain on the computer.

System Architecture-

Input

|

|

To the function named="get_spending"
For(income , expenses)

|

|

To the calculation section that to
Get (total expenses , balance)

|

|

Tabular report

Workflow Diagram-

Start

|

Enter the monthly income of the user.

|

Enter the expense till the loop stop, it will stop with user type “done”.

|

Calculate the total spending that has been done.

|

Find out the remaining balance with help of total spending and income.

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Display the table that contain all the information above mentioned.

|

End

UML Use Case Diagram-

user

|

|

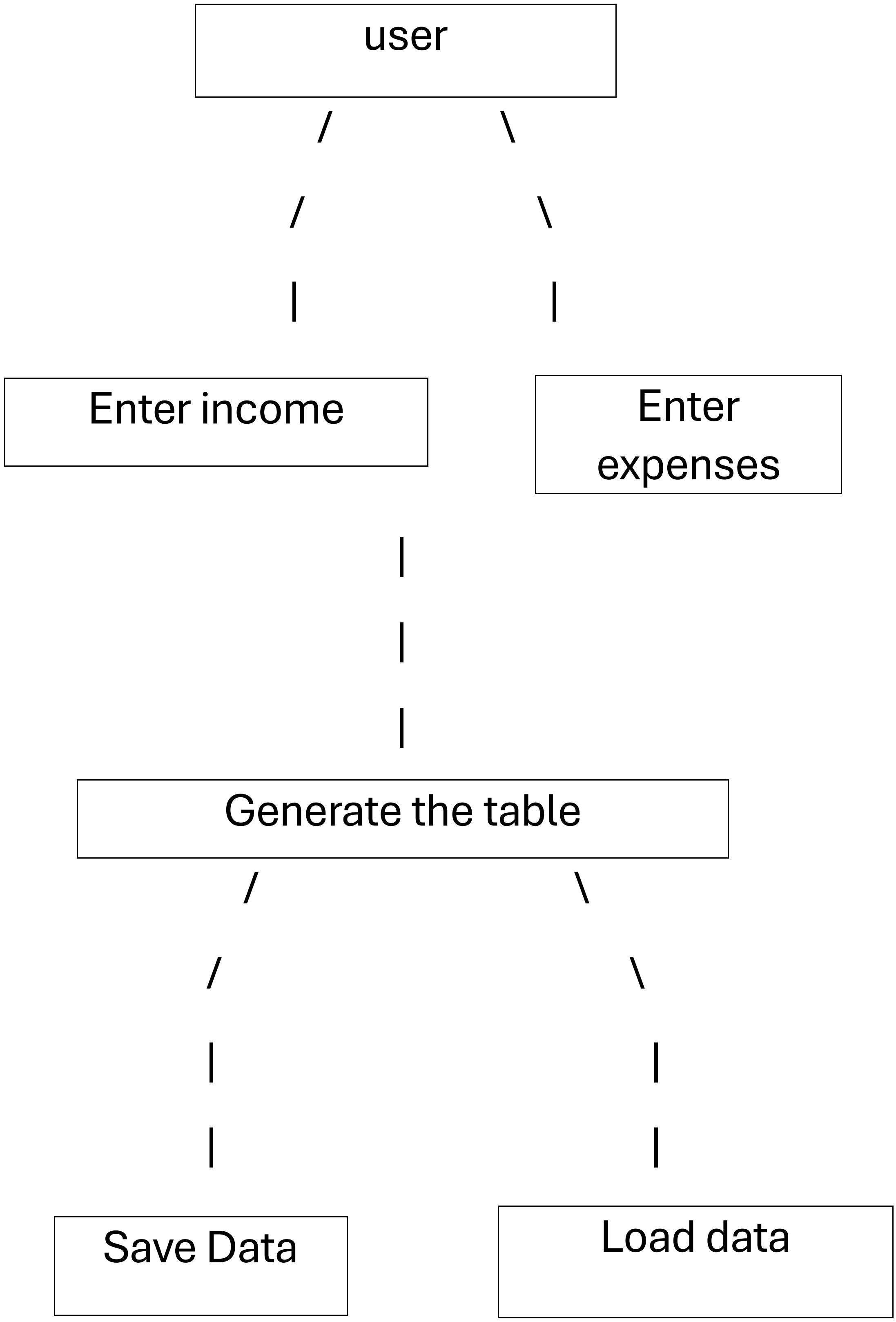
|

V

Budget Report

- Enter income.
- Enter expenses.
- Do all the calculation.
- Generate the report.

Use Case Diagram-



Implementation Details-

So now the program is divided into 3 parts:

1. A function named body:

   Contain all the print statement and control the program flow.

2. A function named spending_money:

   Contain all the input given by the user about where the money spends and the amount then store it in the nested list.

3. There is an if statement that compare the name the body if that’s true then it calls the body function.

Screenshots / Results-

```
========== BUDGET TRACKING SYSTEM ==========
Enter your monthly income (₹): 150000

Enter each spending you made.
When you are done, simply type: done

Where did you spend money?: rent
How much did you spend there?: 20000

Where did you spend money?: groceries
How much did you spend there?: 5286

Where did you spend money?: electricity
How much did you spend there?: 1450

Where did you spend money?: online shopping
How much did you spend there?: 4300

Where did you spend money?: fuel
How much did you spend there?: 1800

Where did you spend money?: done

**************** BUDGET REPORT ****************
Monthly Income: ₹150000

Spending Category            Amount
---------------------------------------------
rent                          20000
groceries                      5286
electricity                    1450
online shopping                4300
fuel                           1800
---------------------------------------------
Total Expenses                32836
Remaining Balance            117164
******************** END ********************
```

Testing Approach-

1. Verify that the program successfully accepts user inputs for income and expenses.

2. Ensure accurate calculations (total expenses, remaining balance).

3. Validate that tabular output is displayed correctly.

4. Confirm that data handling (saving and loading files, if included) works correctly.

5. Check that invalid inputs are handled safely with proper error messages.

6. Ensure the program meets user requirements and is easy to use.

Challenges Faced-

While making this program there are several challenges I need to face some of the challenges are as follows:-

1. I need to create a program that is as small as possible and have less time complexity.
2. Using function to separate the input and output ,how to link them so that when the body function work it will automatically call the spendine_money function.
3. I need the output in tabular form for that I have make the whole bunch of print statement in a specific order using (.format) and (:< , :>) for specific gap.
4. At the last to get the amount and type of spending out of the spending_money for total expenses calculation.

Future Enhancements-

1. Export report to CSV or PDF.
2. Add graphs for expense visualization.
3. Create a GUI using Tkinter.
4. Add login/user accounts.
5. Monthly history tracking.

References -

I got this idea from a hotel receipt that shows the total billing amount , credit card details , room no. ,etc. show me that people can spend there money without making a good use of it.

The structure of my program is inspired from a previous program of a slot machine game that I have build to practice the function in python.

At the end I used my professor ppt for other information relation to the several quarry one of them are how to print the text with maintaining the same gap within them.